Antidote: First Targeted Defenses Against Truth Serum Privacy Attacks

Bruna Vasconcelos

CS 6958: Machine Learning Security — Fall 2025

Abstract

The Truth Serum attack (Tramèr et al., CCS 2022) enables targeted privacy attacks against machine learning models by poisoning training data with mislabeled copies of target samples. This attack amplifies membership inference vulnerabilities by orders of magnitude, yet the original authors explicitly left defenses as an open problem. We propose **Antidote**, the first targeted defense against Truth Serum, which exploits the attack's fundamental mechanism: the insertion of near-duplicate samples with inconsistent labels. Using pretrained ImageNet features to detect these duplicates before training corruption occurs, Antidote achieves a detection F1-score of 0.941 with perfect recall, catching all 2,000 poison samples while generating only 250 false positives. In contrast, Spectral Signatures—a standard backdoor defense—achieves an F1-score of just 0.01, demonstrating that existing defenses are ineffective against label-flipping attacks. Our end-to-end evaluation shows Antidote reduces the attack's privacy amplification by 52.9% while maintaining model utility. We provide comprehensive ablation studies across similarity thresholds and attack configurations, establishing a foundation for defending against privacy-amplifying data poisoning attacks.

1 Introduction

Machine learning models trained on user data are vulnerable to privacy attacks that infer whether specific individuals' data was used in training. Membership inference attacks (MIA) exploit differences in model behavior between training and test samples to make this determination [2]. While standard membership inference achieves limited success against well-regularized models, the Truth Serum attack [1] dramatically amplifies this threat through data poisoning.

1.1 The Truth Serum Attack

Truth Serum enables an adversary to *target specific individuals* for privacy violation. The attack mechanism is deceptively simple: insert k mislabeled copies of each target sample into the training data. For a target image of a cat with true label "cat," the attacker inserts 8 copies labeled "dog." After training, the model exhibits conflicting behavior on the target—it has learned both the correct and incorrect associations—resulting in elevated loss that makes the target trivially identifiable via membership inference.

The original paper demonstrated attack success rates improving from 7% to 59% at 0.1% false positive rate, and noted that poisoning eliminates the need for expensive shadow model training. The authors explicitly acknowledged:

> *"We leave the development of targeted defenses against our attack as an open problem for future work."* — Tramèr et al., Section 7

This project directly addresses that open problem.

1.2 Research Question

Can we detect and mitigate Truth Serum data poisoning attacks by exploiting statistical signatures inherent to the attack mechanism?

1.3 Contributions

1. **First targeted defense** against Truth Serum attacks, achieving 0.941 F1-score on poison detection
2. **Principled approach** exploiting the attack's unavoidable signature: near-duplicate samples with label mismatches
3. **Comprehensive evaluation** including comparison against Spectral Signatures baseline and ablation studies
4. **End-to-end validation** demonstrating 52.9% reduction in attack effectiveness

2 Background and Related Work

2.1 Membership Inference Attacks

Membership inference attacks determine whether a specific sample was in a model's training set [2]. Standard attacks exploit the observation that models exhibit lower loss (higher confidence) on training samples due to memorization. However, these attacks are often ineffective at low false positive rates against properly regularized models.

2.2 Truth Serum Attack Mechanism

The Truth Serum attack [1] inverts the standard MIA assumption. Instead of relying on low loss to identify members, the attack *creates* high loss on targets through poisoning:

1. Select target samples (x_t, y_t) for privacy attack
2. Generate poison samples (x_t, y') where $y' \neq y_t$
3. Insert k copies of each poison into training data
4. After training, targets exhibit elevated loss due to label conflict
5. High loss becomes the membership signal

This attack is particularly concerning for Machine Learning as a Service (MLaaS) platforms where users contribute training data.

2.3 Existing Defenses and Their Limitations

Spectral Signatures [3] detects backdoor attacks by identifying outliers in the covariance spectrum of learned representations. However, this assumes poisoned samples are *feature-space outliers*. In label-flipping attacks, features remain unchanged—only labels are modified—making spectral methods ineffective.

Activation Clustering [4] clusters neural network activations to identify anomalous samples. This defense targets trigger-based backdoors that create distinct activation patterns, not mislabeled samples with normal activations.

STRIP [5] detects trojans at inference time by measuring prediction entropy under perturbation. As an inference-time defense, it cannot prevent training-time poisoning.

Differential Privacy [6] bounds information leakage but does not prevent poisoning—it only limits the influence of individual samples.

None of these defenses target the specific mechanism of label-flipping attacks. Our work fills this gap.

3 Threat Model

3.1 Adversary

- **Goal:** Infer membership of specific target samples with high confidence
- **Knowledge:** Knows target samples (x, y); does not know full training set
- **Capabilities:** Can contribute data to training pool; can query trained model
- **Strategy:** Insert k mislabeled copies of each target

3.2 Defender (Our Work)

- **Goal:** Detect and remove poisoned samples before training
- **Knowledge:** Full training dataset $\mathcal{D}'$; no knowledge of which samples are poisoned

- **Capabilities:** Can analyze and filter training data; access to pretrained models
- **Constraints:** Must preserve model utility (accuracy drop $< 10\%$)

3.3 Formal Definition

Let $\mathcal{D}$ be the clean training set and $\mathcal{P} = \{(x_t, y')\}_{t \in T}$ be the poison set where $y' \neq y_t$.

Attack: $\mathcal{D}' = \mathcal{D} \cup \mathcal{P}$

Defense: Find $\hat{\mathcal{P}} \subseteq \mathcal{D}'$ such that $\hat{\mathcal{P}} \approx \mathcal{P}$, then train on $\mathcal{D}' \setminus \hat{\mathcal{P}}$

4 Proposed Method: Antidote

4.1 Key Insight

The Truth Serum attack's effectiveness relies on a fundamental constraint: *the attacker must insert near-duplicate copies with different labels*. This creates a detectable signature that the attacker cannot avoid—the attack's mechanism is also its weakness.

4.2 Critical Design Decision: Pretrained Features

A naive approach would use the model being trained to extract features for duplicate detection. However, this creates a circular dependency: the model's representations are corrupted by the poisoning we're trying to detect.

Our key innovation is using **pretrained ImageNet features** from a ResNet-18 model that has never seen the (potentially poisoned) training data. This provides clean feature representations independent of any poisoning, breaking the circular dependency.

4.3 Algorithm

The algorithm (Algorithm 1) operates in two phases:

Phase 1 extracts 512-dimensional features from all samples using an ImageNet-pretrained ResNet-18 with the classification head removed. Features are L2-normalized for cosine similarity computation.

Phase 2 identifies suspicious samples through two criteria:

Algorithm 1 Antidote Defense

Require: Training set $\mathcal{D}'$, threshold τ, pretrained model f_θ
Ensure: Filtered dataset $\mathcal{D}_{clean}$

Phase 1: Feature Extraction
for each $(x_i, y_i) \in \mathcal{D}'$ **do**
  $z_i \leftarrow f_\theta(x_i)$ ▷ 512-dim features
  $\hat{z}_i \leftarrow z_i / \|z_i\|_2$ ▷ Normalize
end for
Phase 2: Near-Duplicate Detection
$S \leftarrow \emptyset$ ▷ Suspected poisons
for each sample i **do**
  $\text{sims}_i \leftarrow \hat{z}_i^\top \hat{Z}$ ▷ Cosine similarities
  $N_i \leftarrow \{j : \text{sims}_{ij} > \tau, j \neq i\}$ ▷ Near-duplicates
  if $|N_i| > 0$ **and** $\exists j \in N_i : y_j \neq y_i$ **then**
    $S \leftarrow S \cup \{i\}$ ▷ Label mismatch
  end if
  if $|N_i| \geq 3$ **then**
    $S \leftarrow S \cup \{i\}$ ▷ Suspicious cluster
  end if
end for
return $\mathcal{D}_{clean} \leftarrow \mathcal{D}' \setminus S$

1. **Label Mismatch:** Sample has a near-duplicate (similarity $> \tau$) with a different label
2. **Cluster Size:** Sample has 3+ near-duplicates (indicates poison copy cluster)

Both criteria are necessary: label mismatch catches the poison copies themselves, while cluster size catches edge cases where all copies happen to have the same wrong label.

4.4 Why This Works

Consider a target cat image. After poisoning:

- Original: $(x_{cat}, \text{“cat”})$ in dataset
- Poisons: 8 copies of $(x_{cat}, \text{“dog”})$ added

All 9 samples (1 original + 8 poisons) have nearly identical features since they're the same image. However, they have inconsistent labels. Our algorithm detects:

- Each poison copy has duplicates with mismatched labels (the original)

Table 1: Detection Performance

Method	Precision	Recall	F1
Antidote (Ours)	0.889	**1.000**	**0.941**
Spectral Signatures	0.009	0.012	0.010

- The original has duplicates with mismatched labels (the poisons)
- All 9 samples belong to a cluster of size ≥ 3

This results in all poison copies being flagged, plus the original target—a conservative approach that may slightly reduce training data but ensures complete poison removal.

5 Experimental Evaluation

5.1 Setup

Dataset: CIFAR-10 (50,000 training, 10,000 test samples)

Model: ResNet-18 adapted for 32×32 images (modified first conv layer, no max pooling)

Attack Configuration: 250 targets × 8 copies = 2,000 poison samples (3.8% of augmented training set)

Training: SGD with momentum 0.9, learning rate 0.1, cosine annealing, 25 epochs

Compute: Google Colab with A100 GPU

5.2 Detection Performance

Table 1 presents detection metrics across three experimental conditions.

Antidote achieves perfect recall (1.000), catching all 2,000 poison samples, with precision of 0.889 (250 false positives out of 50,000 clean samples). The F1-score of 0.941 represents near-optimal detection.

Critically, Spectral Signatures achieves an F1-score of just 0.010—effectively random performance. This validates our hypothesis that backdoor defenses designed for trigger-based attacks are ineffective against label-flipping attacks where features remain unchanged.

Table 2: Similarity Threshold Ablation

Threshold	Precision	Recall	F1
0.95	0.820	1.000	0.901
0.99	0.889	1.000	0.941
0.999	0.889	1.000	0.941

Table 3: Number of Copies Ablation

Copies (k)	Precision	Recall	F1
2	0.667	1.000	0.800
4	0.800	1.000	0.889
8	0.889	1.000	0.941
16	0.941	1.000	0.970

5.3 Ablation Studies

5.3.1 Similarity Threshold

Table 2 shows detection performance across threshold values.

Lower thresholds (0.95) increase false positives as more legitimate similar samples are flagged. Thresholds of 0.99 and 0.999 perform identically, suggesting 0.99 is sufficient for detecting exact duplicates while being robust to minor variations.

5.3.2 Number of Poison Copies

Table 3 shows robustness across attack configurations.

Detection improves with more copies, as larger clusters are more distinctive. Even with only $k = 2$ copies, Antidote achieves F1=0.800 with perfect recall. This graceful degradation means the defense remains effective even if attackers reduce their poison budget.

5.4 End-to-End Defense Evaluation

Table 4 presents the complete pipeline evaluation.

Attack Verification: The poisoned model shows 22× elevated target loss (0.027 → 0.593), confirming Truth Serum's mechanism. MI AUC increases from 0.497 to 0.546, representing successful attack amplification.

Table 4: End-to-End Defense Effectiveness

Model	Test Acc	MI AUC	Target Loss
Clean	92.1%	0.497	0.027
Poisoned	83.5%	0.546	0.593
Defended	85.0%	0.520	0.561

Defense Effectiveness: Antidote reduces MI AUC from 0.546 to 0.520, achieving 52.9% reduction of the attack's privacy amplification. The defended model maintains 85.0% accuracy, only 7.1% below the clean baseline.

5.5 Loss Distribution Analysis

Figure **??** visualizes the loss distributions that underpin these results. The clean model exhibits near-zero loss on targets (mean 0.027), as expected for well-fit training samples. The poisoned model shifts target losses dramatically higher (mean 0.593), creating the separation that enables membership inference. The defended model partially restores normal behavior.

6 Discussion

6.1 Why Spectral Signatures Fails

Spectral Signatures identifies outliers in the learned representation space. This works for trigger-based backdoors because trigger patterns create distinctive activations. However, Truth Serum poisons are *visually identical* to legitimate samples—only their labels differ. In feature space, a cat labeled "dog" looks exactly like a cat labeled "cat." There is no spectral anomaly to detect.

This fundamental mismatch explains the 0.01 F1-score: Spectral Signatures flags samples based on feature-space outlier scores, which are uncorrelated with label-flipping poisoning.

6.2 Limitations

Accuracy Cost: The defended model shows 7.1% accuracy drop from clean baseline. This occurs because we remove the original target samples along with poisons (they're part of the same duplicate cluster). Future work could explore more surgical removal.

Moderate Attack Amplification: Our experiments show MI AUC of 0.546, lower than the >0.99 reported in the original paper. This may be due to shorter training (25 vs. 50+ epochs), smaller model capacity, or differences in attack evaluation methodology. Despite this, the attack mechanism is clearly visible in loss distributions.

Adaptive Attackers: An adversary aware of Antidote could attempt countermeasures:

- *Adding noise to copies:* Would reduce cosine similarity below threshold. However, sufficient noise to evade detection would also weaken the attack's effectiveness.
- *Using fewer copies:* Our ablation shows detection remains effective even at $k = 2$.

6.3 Broader Implications

This work demonstrates that defenses must be *mechanism-specific*. The failure of Spectral Signatures against Truth Serum, despite its effectiveness against trigger-based backdoors, illustrates the importance of understanding attack mechanisms when designing defenses. Our approach of exploiting the attack's unavoidable constraints provides a template for future defense development.

7 Conclusion

We presented Antidote, the first targeted defense against Truth Serum privacy attacks. By exploiting the attack's fundamental requirement—inserting near-duplicate samples with inconsistent labels—we achieve near-perfect detection (F1=0.941) while demonstrating that existing backdoor defenses are ineffective. Our comprehensive evaluation across thresholds, attack configurations, and end-to-end metrics establishes a foundation for defending against privacy-amplifying data poisoning.

7.1 Future Work

- Extend evaluation to larger datasets (ImageNet) and model architectures

- Develop adaptive attack variants and corresponding defenses
- Explore integration with differential privacy for layered protection
- Investigate detection during training rather than as preprocessing

References

[1] F. Tramèr, R. Shokri, A. San Joaquin, H. Le, M. Jagielski, S. Hong, and N. Carlini. Truth Serum: Poisoning Machine Learning Models to Reveal Their Secrets. In *ACM CCS*, 2022.

[2] R. Shokri, M. Stronati, C. Song, and V. Shmatikov. Membership Inference Attacks Against Machine Learning Models. In *IEEE S&P*, 2017.

[3] B. Tran, J. Li, and A. Madry. Spectral Signatures in Backdoor Attacks. In *NeurIPS*, 2018.

[4] B. Chen, W. Carvalho, N. Baracaldo, H. Ludwig, B. Edwards, T. Lee, I. Molloy, and B. Srivastava. Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering. In *NDSS*, 2019.

[5] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal. STRIP: A Defence Against Trojan Attacks on Deep Neural Networks. In *ACSAC*, 2019.

[6] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang. Deep Learning with Differential Privacy. In *ACM CCS*, 2016.

[7] C. G. Northcutt, L. Jiang, and I. L. Chuang. Confident Learning: Estimating Uncertainty in Dataset Labels. *JAIR*, 70:1373–1411, 2021.